

Project Workflow: Week 4

Phase: UI Layout & Game Window Development (Bounce Ball Game)

1. Objective

The primary goal of Week 4 was to design and implement the complete visual layout and user interface for the Bounce Ball Game. This included:

- Developing the main game window and primary play area panel.
- Creating the score display and game status information area.
- Basic visual styling to the GUI.
- Designing the game canvas layout for paddle, ball, and brick/boundary elements.
- Preparing the graphical interface for ball physics and user input integration.

2. Detailed Task Breakdown

Main Game Window & Play Area Panel Development

Developed the core game window using JFrame and constructed the primary play area panel using JPanel. Configured optimal dimensions, fixed layout boundaries, and double-buffering to ensure smooth gameplay graphics.

Score Display & Status Area Creation

Designed a status text area to show start, pause, and game-over messages clearly, and counting score.

Theme & Basic GUI Styling Application

Applied a modern, clean visual theme across all graphical components. Selected custom color palettes for the canvas background, paddle, bouncing ball, and status indicators .

3. Deliverables

- **Main Game Window:** Fully configured Java Swing JFrame window for the Bounce Ball Game.
- **Play Area Panel:** Functional JPanel play area canvas designed for bounce physics rendering.
- **Score Display:** Integrated score .
- **Status Area:** Real-time game status banner for game start, pause, and game-over notifications.
- **GUI Visual Theme:** Consistent colors, antialiased graphics rendering, and clear fonts applied.

4. Tools & Resources

Development Environment

- Java JDK 11+
- IntelliJ IDEA / VS Code / Eclipse

Libraries & Technologies

- Java Swing (javax.swing)
- Java AWT (java.awt)
- JFrame & JPanel
- Graphics & Graphics2D